Ben Gilder

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Second year mathematics student at the University of Bath, with experience in mathematical programming, software development and educational support. Contributed to the TBRe Formula Student team, implementing custom GUI and data solutions.

Education

Mathematics BSc (Professional Placement) | University of Bath Sep 2025 – Jun 2029

End of 1st Year Results – **First Class (Average 87%)**

Matthew Arnold School Sep 2017 – Jun 2024

A Levels (2024) – Further Mathematics **A***, Physics **A***, Chemistry **A**, Mathematics **A**

GCSEs (2022) – 11 at Grades **9-7** with Mathematics **(9)** and English **(9)**

Technical Skills

Languages – Python, SQL, R, JavaScript, C#, C

Frameworks and Tools – NumPy, Git, GitHub, LaTeX, WPF

Projects

Tensor Library Implementation | GitHub Project Aug 2026

- Implemented a fully featured Tensor library from basic C, with 18 different functions like permutations, additions and multiplication
- Handled memory allocation manually, and hid the internals of the tensor constructs from outside the library, only exposing an opaque pointer for less risk of unintended behaviour
- Wrote a comprehensive README explaining how to use the library and how it behaves

Custom GUI and Driver Integration | TBRe Formula Student Oct 2025 – Feb 2026

- Prioritised low-latency data visualisation through the use of specific Vector hardware protocols
- Developed a custom Windows application using the WPF framework in C#
- Collaborated with a multidisciplinary team to integrate the solution

Experience

Lego Robotics Tutor | Code Cadets Sep 2024 – Jan 2025

- Managed STEM activities for up to 30 children from ages 4-11
- Instructed students in STEM activities using the SPIKE prime lego sets, with block coding and Python
- Managed and handled high value company equipment

Learning Support Assistant | Matthew Arnold School Sep 2024 – Feb 2025

- Supported students with learning difficulties in both emotional and accessibility aspects
- Created and maintained an organised digital file system for class notes
- Utilised specialist assistive technology to improve lesson engagement for students with specialist learning needs

Software Development Internship | Soft Pauer Jul 2022 – Aug 2022

- Developed and implemented custom SQL and Javascript solutions within the company's data systems
- Collaborated with senior engineers to provide efficient and effective integration solutions

University Courses

Linear Algebra - 86% Sep 2025 - May 2026

- Covered the abstract mathematical concepts of vector spaces, inner product spaces, linear maps, determinants and fields
- Applied these abstract ideas to create algorithms for applied functions, like determinant calculation and general matrix inversion

Analysis - 91% Sep 2025 - May 2026

- Formed the system of real numbers from axioms, leading to higher level concepts like differentiation, convergence and divergance, and infinite sums
- Studied theorems and results to prove the existence of solutions to generalised problems

Vector Calculus and Differential Equations - 90% Sep 2025 - May 2026

- Studied the generalisation of calculus topics to higher dimensions, while covering topics like line gradients, curvature, normals, and orbital equations
- Learned to solve a wide variety of ODEs in higher dimensions, for application to real systems

Probability and Statistics - 92% Sep 2025 - May 2026

- Studied and manipulated datasets through the use of R
- Typeset reports in LaTeX, with rigorous mathematical explanation of results

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Self-Study

Fundamentals and Workings of C | Beej's Guide to C Programming Jun 2026

- Covered language functions like pointers, macros, compiler details, structs, memory allocation, and memory safety
- Applied this to projects implementing mathematical objects like tensors and matrices, with robust error detection and memory handling

Markov Chains | J R Norris Jun 2026

- Studied the chapters on discrete Markov Chains, covering topics like hitting times, absorbtion probabilities, communicating classes and more
- Worked through the given examples and problems to gain a practical knowledge of how to apply the theory

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Awards and Achievements

Cambridge Battlecode – Reached 5th for Python out of 200 international competitors
UKMT Maths Challenge – Multiple Gold Awards and Entries to the Second Round Kangaroo
BUCS Silver Medal for Judo – Won the silver medal in the novice division
Gold Medal for Judo Bath Competition – Won the gold medal in the novice division in an internal Bath University competition
Bath University Climbing Team – Placed on the University of Bath's climbing team and competed in BUCS on a national level
Duke of Edinburgh – Bronze (2021) and Silver (2022)

References

Yury Korolev – Academic Advisor – *ymk30@bath.ac.uk*
Alexander Hurst - A Level Maths Teacher - *ahurst@maschool.org.uk*